Zhenyu (Kevin) Zhu

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EDUCATION

University of Chicago **Chicago, USA**

- Master of Science in Applied Data Science (MS-ADS) Sep 2026 - May 2028 (Expected)

Virginia Tech **Blacksburg, USA**

- B.S. in Business Administration in Entrepreneurship, Innovation and Technology Management Sep 2022-Jun 2026
- GPA: 3.8/4.0

Xidian University **Xi'an, CN**

- B.Mgt. in Big Data Management and Application Sep 2022-Jun 2026
- GPA: 3.8/4.0 | Awards: 2023 School Scholarship (Top 6%), Xidian University

EXPERIENCE

Bringspring Science and Technology Co., Ltd, Zhengzhou Headquarters **Zhengzhou, CN**

Research & Development Intern Jun - Aug 2025

- Designed MySQL schemas for 10+ operational systems, including staffing, materials, and fixed assets, and supported Apache Flink streaming ingestion; built DBeaver test databases to validate data quality before release.
- Built 10+ SQL operational KPIs and Datart dashboards to monitor work-order, security, and wastewater systems; refactored Python API error handling to improve service reliability.

Orient Securities Co., Ltd **Shanghai, CN**

FOF Investment Research Intern, Securities Investment Business Headquarters Jul - Oct 2024

- Replicated and backtested two broker asset-allocation models (Haitong counter-cyclical, Guosheng six-cycle rotation) in Python using Wind data to support FOF strategy decisions.
- Automated fund-holdings and industry-deviation processing with Python scripts and authored research reports for pre-investment analysis.

Bank of Guangzhou **Guangzhou, CN**

System Operation and Maintenance Intern, The Technology Department Jan - Feb 2024

- Deployed an ELK log-monitoring stack on 5 Linux VMs and replaced Logstash with lightweight Filebeat agents to reduce ingestion overhead and enable faster, scalable log search.

PROJECTS

Chicago Health Price Benchmarking – Cloud Data Engineering Final Project Oct-Dec 2026 (Expected)

- Co-building (4-person team) a GCP-to-Snowflake pipeline that streams 6 TB of public insurer price-transparency files through Cloud Storage and Cloud SQL into a Snowflake star schema with SCD2 history and reconciled, re-runnable loads.
- Leading analysis and serving: exploratory analysis of negotiated rates across insurers, networks, and Chicago facilities; a Streamlit dashboard for employers and employees; and Claude-powered exploration via an MCP server.

Construction of Tourism Association Map of Xi'an City based on Data Mining Oct 2025-June 2026

Graduation Thesis, supervised by Dr. Lei Sun, Xidian University

- Processed 1,000 Ctrip travel notes through text cleaning, entity extraction, synonym normalization, and screening, retaining 477 valid analytical samples.
- Applied FP-Growth and PrefixSpan-inspired sequential mining to produce 335 association rules and 1,448 sequential patterns; deployed a Neo4j graph with 35 nodes and 460 directed relationships.

2026 MCM/ICM Problem D - WNBA Team Management Optimization (Honorable Award) 2026

- Led Python coding, data processing, and visualization for an integer-programming roster optimizer under a salary cap.
- Ran scenario and sensitivity analysis projecting 30–40% team valuation upside from adding a second superstar.

Large Language Models Online Research Seminar

Research Participant, supervised by Dr. Pavlos Protopapas, Harvard University June 2025-Aug 2025

- AI Teaching Assistant Chatbot, UnivAI Platform — Built a RAG-based assistant with LangChain, Haystack, reranking, and Streamlit to retrieve course materials and answer recurring student queries.
- Graph-Rationale-Guided RAG for Medical QA — Developed a GRAG framework combining UMLS-Neo4j knowledge graphs, multi-source retrieval, hallucination detection, and QLoRA.
- Evaluated on MedQA, MedMCQA, and RareDisease-MedQuAD, achieving 10–12% higher accuracy, ~20% lower hallucination rates, and >80% graph fidelity; published in Academic Journal of Science and Technology, Vol. 19, No. 2, 2026, pp. 321–331.

SKILLS

- Programming & Analytics: Python, SQL, MySQL, Excel
- AI / Machine Learning: LLMs, RAG, PyTorch, Transformers, QLoRA, Knowledge Graphs, Neo4j, LangChain, Haystack, MCP
- Cloud & Data Engineering: GCP (Cloud Storage, Cloud SQL), Snowflake, Apache Flink, ELK, Filebeat, Streamlit, Datart